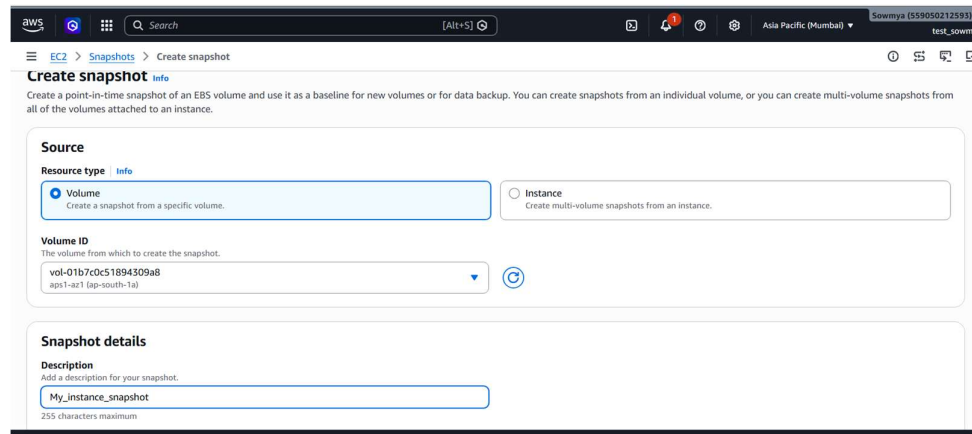


Demonstration of cloud cost optimization

Step 1: Login to your AWS IAM user account

Step 2: Go to EC2 and create an EC2 instance

Step 3: Go to ec2 dashboard->select snapshots and create a snapshot for the ec2 instance volume you have created



Create snapshot [Info](#)

Create a point-in-time snapshot of an EBS volume and use it as a baseline for new volumes or for data backup. You can create snapshots from an individual volume, or you can create multi-volume snapshots from all of the volumes attached to an instance.

Source

Resource type [Info](#)

☒ **Volume**
Create a snapshot from a specific volume.

☐ **Instance**
Create multi-volume snapshots from an instance.

Volume ID
The volume from which to create the snapshot.

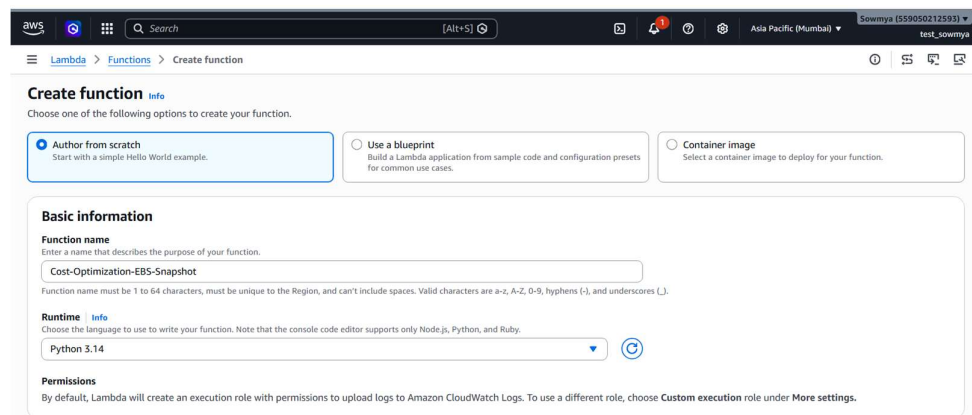
vol-01b7c0c51894309a8
aps1-a21 (ap-south-1a)

Snapshot details

Description
Add a description for your snapshot.

My_instance_snapshot
255 characters maximum

Step 4: Go to Lambda functions in AWS console and create a lambda function



Create function [Info](#)

Choose one of the following options to create your function.

☒ **Author from scratch**
Start with a simple Hello World example.

☐ **Use a blueprint**
Build a Lambda application from sample code and configuration presets for common use cases.

☐ **Container image**
Select a container image to deploy for your function.

Basic information

Function name
Enter a name that describes the purpose of your function.

Cost-Optimization-EBS-Snapshot
Function name must be 1 to 64 characters, must be unique to the Region, and can't include spaces. Valid characters are a-z, A-Z, 0-9, hyphens (-), and underscores (_).

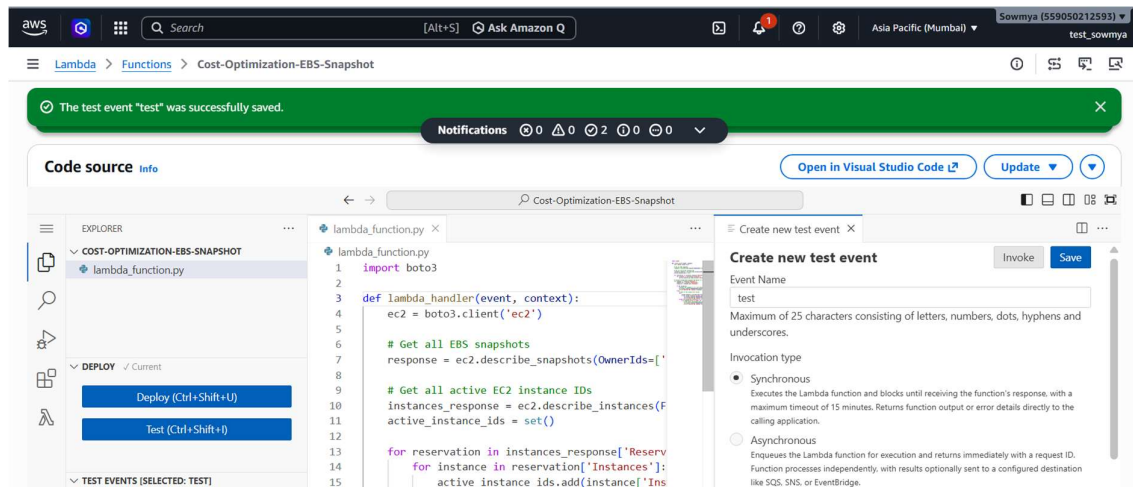
Runtime [Info](#)
Choose the language to use to write your function. Note that the console code editor supports only Node.js, Python, and Ruby.

Python 3.14

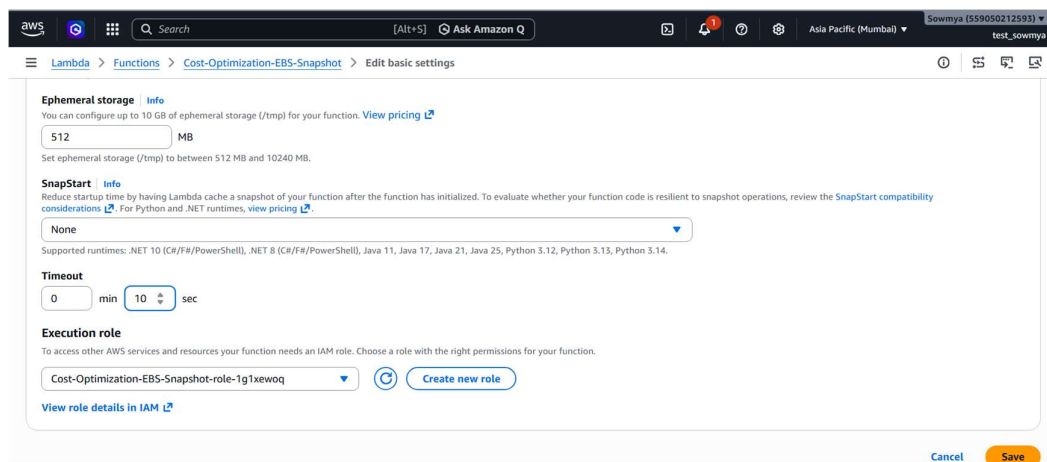
Permissions
By default, Lambda will create an execution role with permissions to upload logs to Amazon CloudWatch Logs. To use a different role, choose **Custom execution role** under **More settings**.

Step 5: Copy lambda function code from below and paste in code source . Then deploy the code and after save. It will ask for event click on create new event as we are manually invoking the event here and then save it.

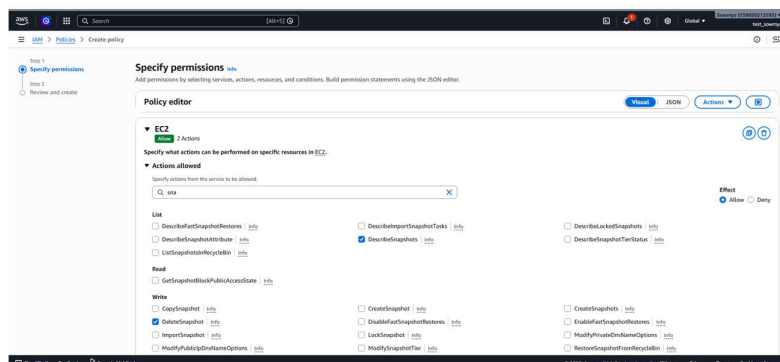
https://github.com/iam-veeramalla/aws-devops-zero-to-hero/blob/main/day-18/ebs_stale_snapshots.py



Step 6: By default lambda function execution time is 3 seconds so the code fails. Change the execution time. Go to configuration tab->click on edit and increase timeout to 10sec. click on save.



Step 7: If you test again it fails and donot given permissions to describe and delete snapshots
Go to configuration tab->Permissions->click on rolename->add permissions.
Here you donot have required snapshot permissions so go to iam and create a policy.



Review and create

Review the permissions, specify details, and tags.

Policy details

Policy name
Enter a meaningful name to identify this policy.

Description - optional
Add a short description for this policy.

Permissions defined in this policy

Permissions defined in this policy document specify which actions are allowed or denied. To define permissions for an IAM identity (user, user group, or role), attach a policy to it.

Allow (1 of 469 services)

Service	Access level	Resource	Request condition
EC2	Limited: LAC, Write	All resources	None

Add tags - optional

Tags are key-value pairs that you can add to AWS resources to help identify, organize, or search for resources.

Step 8: Attach the policy created and test again

Attach policy to Cost-Optimization-EBS-Snapshot-role-1g1xewoq

Current permissions policies (1)

Policy name	Type	Attached entities
AWSLambdaBasicExecutionRole	Customer managed	

Other permissions policies (1/1156)

Filter by Type: All types 1 match

Policy name	Type	Description
cost-optimization-efs-policy	Customer managed	

Step 9: It fails as we did not described the instances and volumes.

Same way create a policy by selecting describevolumes and describeinstances

Policy was successfully attached to role.

Attach policy to Cost-Optimization-EBS-Snapshot-role-1g1xewoq

Current permissions policies (2)

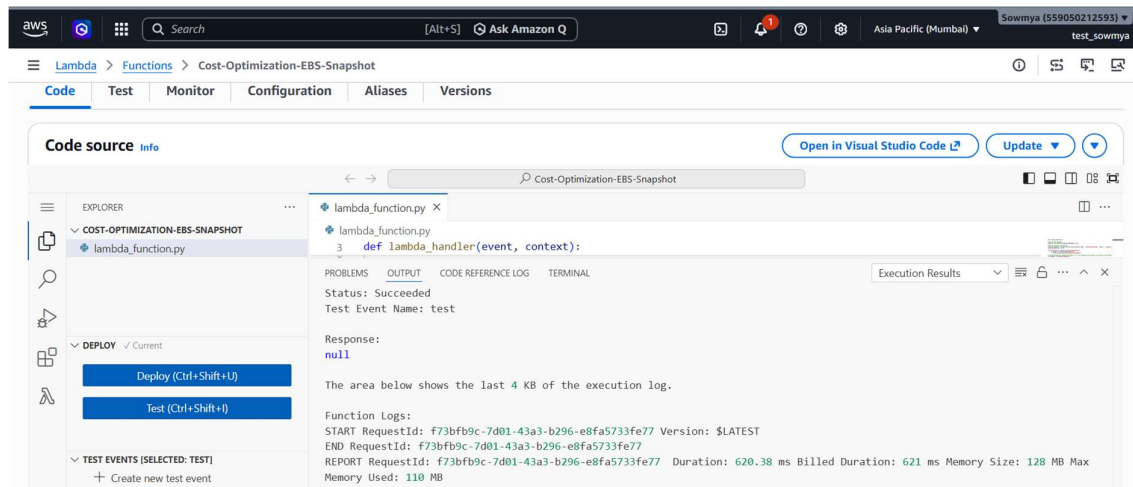
Policy name	Type	Description
ec2-permissions	Customer managed	

Other permissions policies (1/1156)

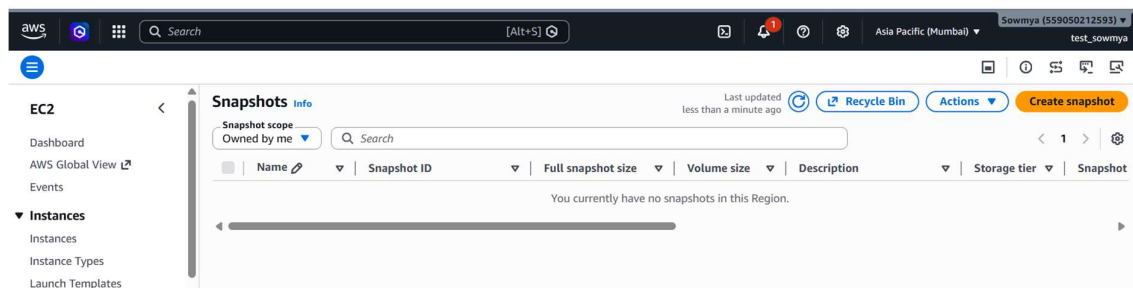
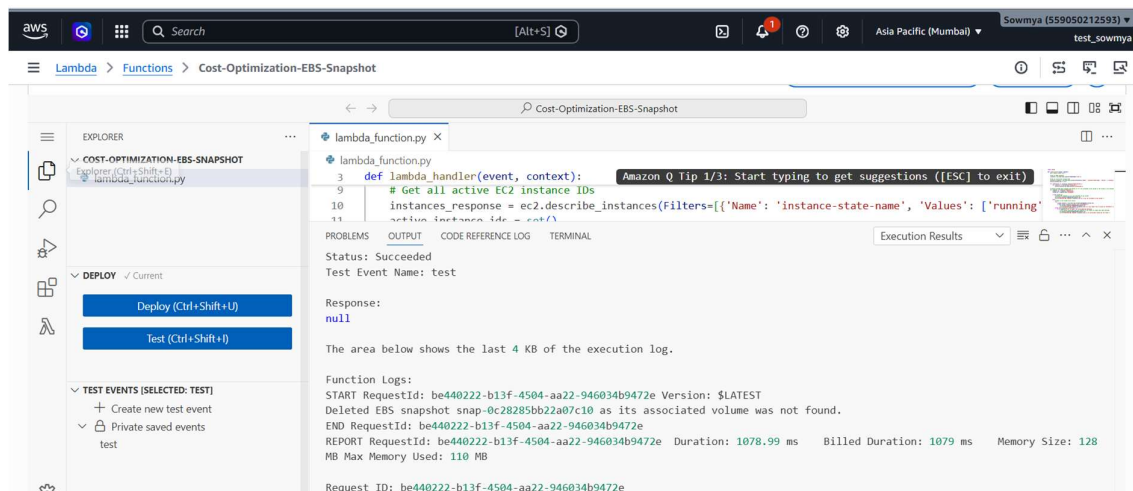
Filter by Type: All types 1 match

Policy name	Type	Description
ec2-permissions	Customer managed	

Now the script got succeeded but it will not delete any snapshots as we have active ec2 and volumes attached.



Step 10: Go to ec2 instance and delete the instance. Then test the lambda function again. It deletes the EBS snapshot created now.



Also it will delete the snapshots if the volumes attached to it are not attached to any ec2 instance.